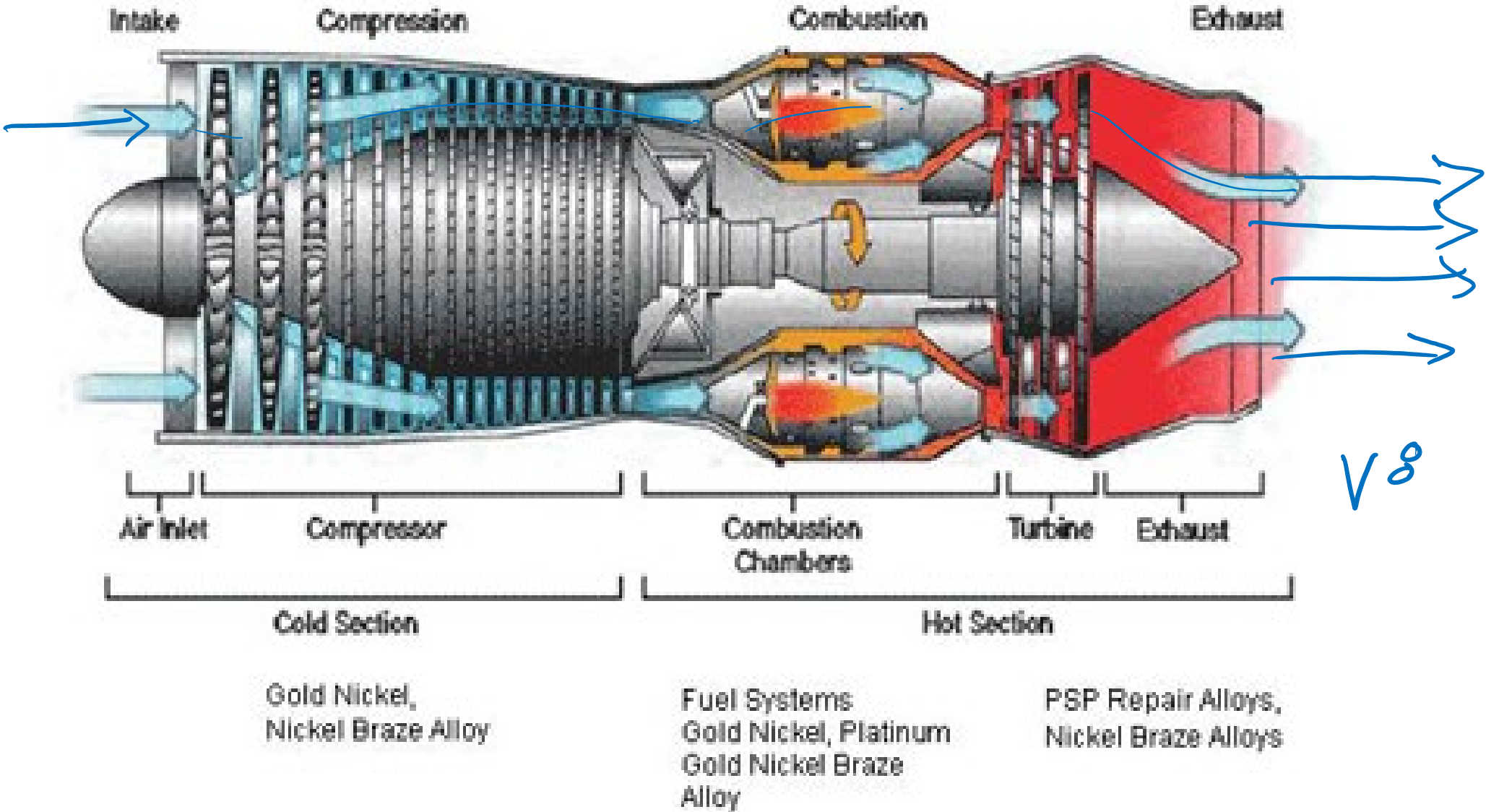




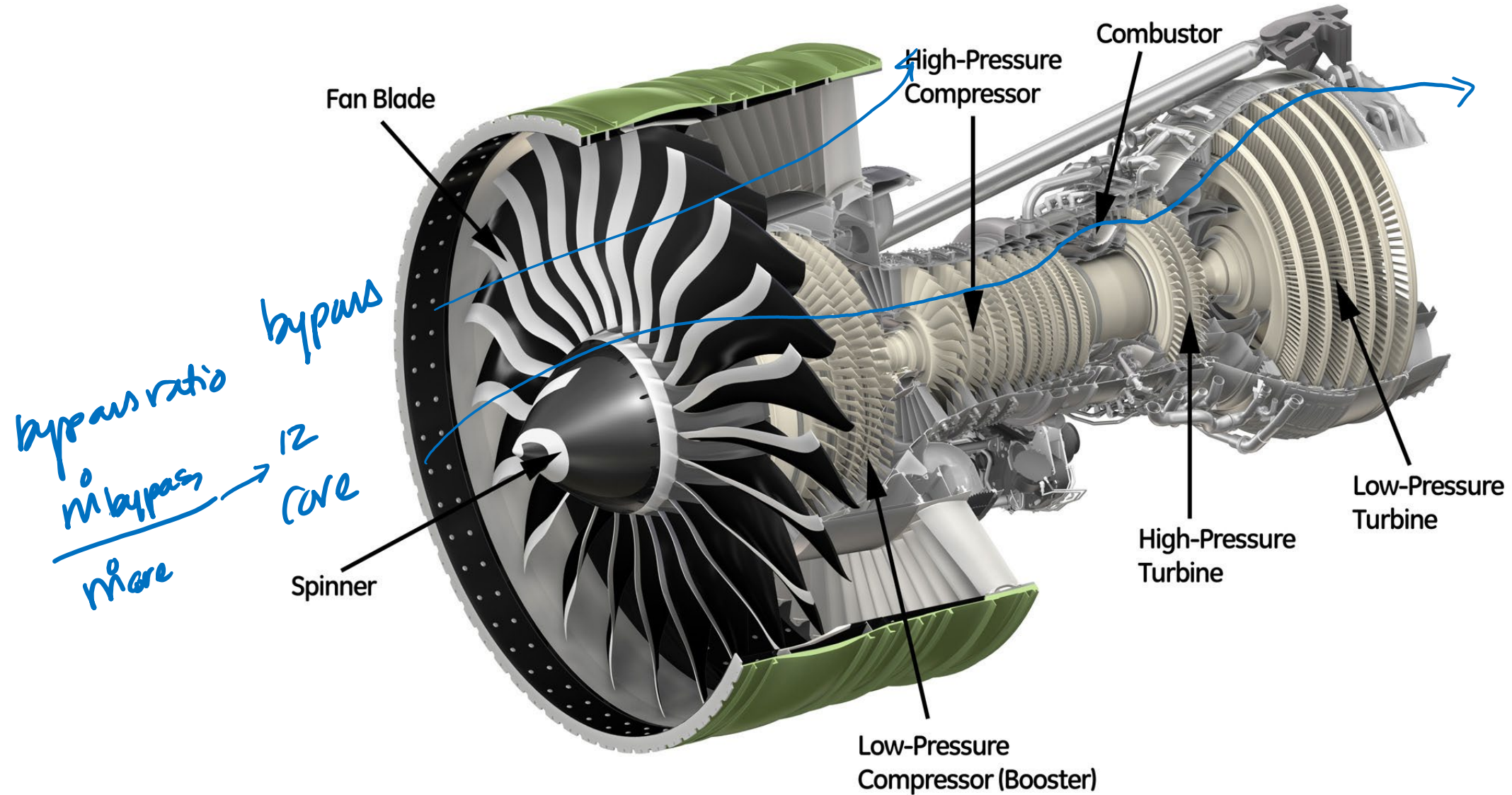
ME 300:
Engineering Thermodynamics

Gas Turbine Engines

Turbojets

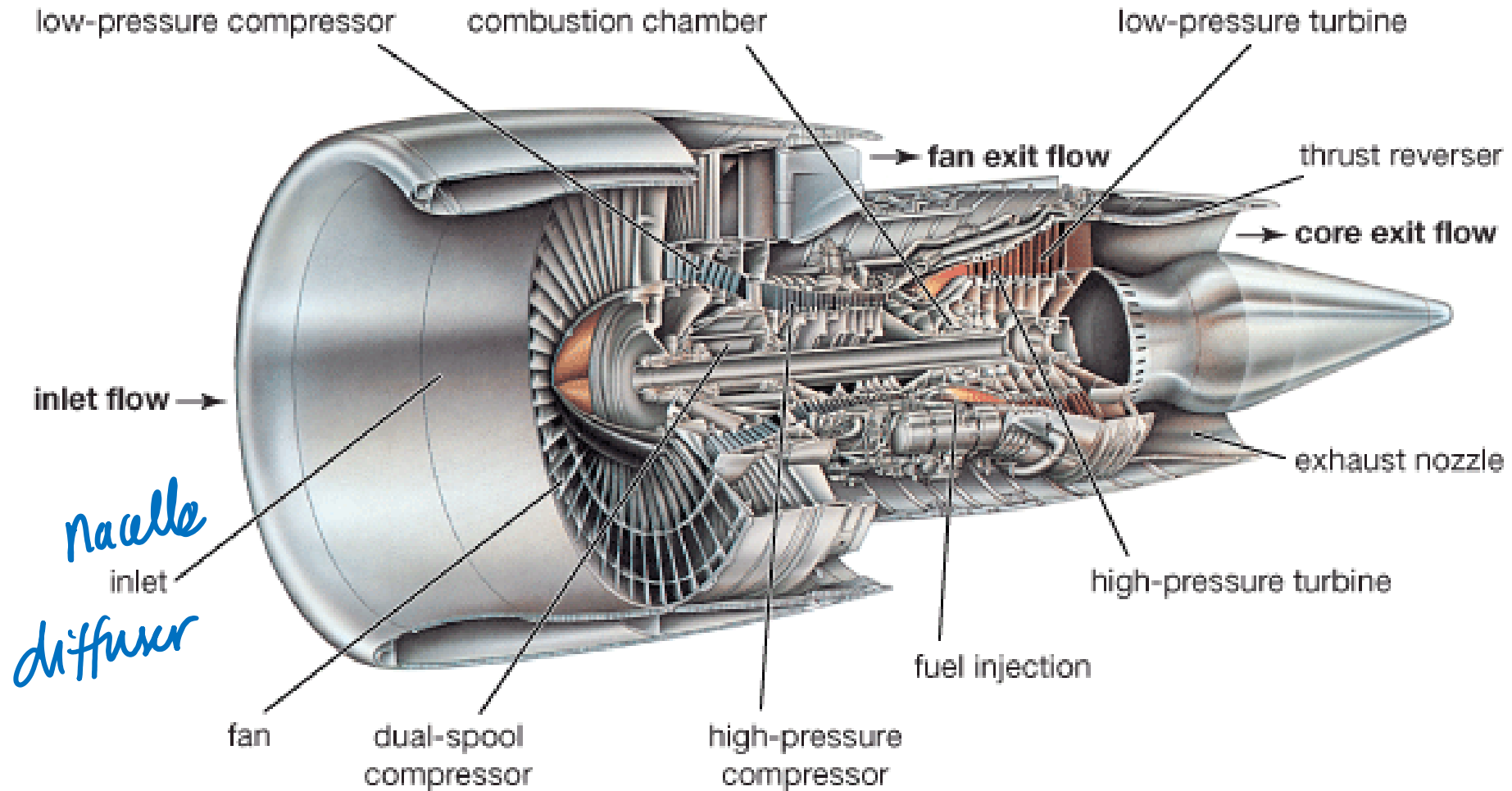


Turbofans



Turbofans

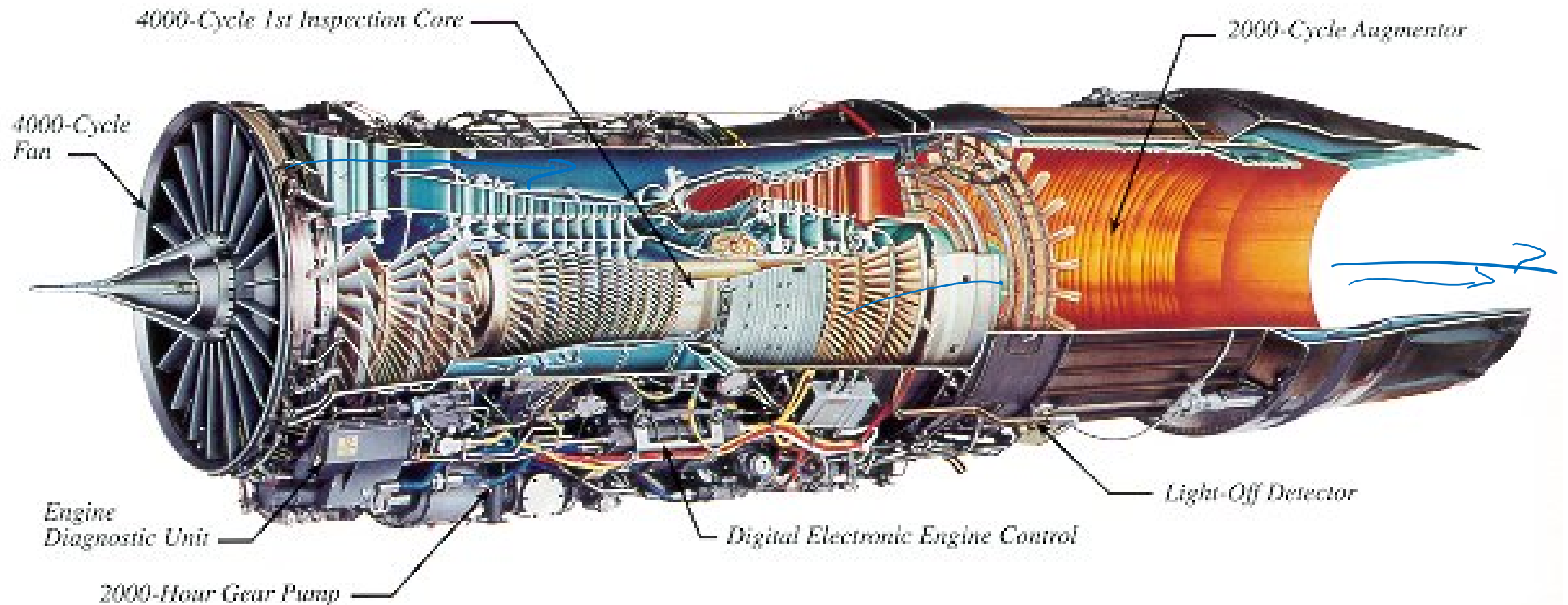
Turbofan jet engine propulsion system



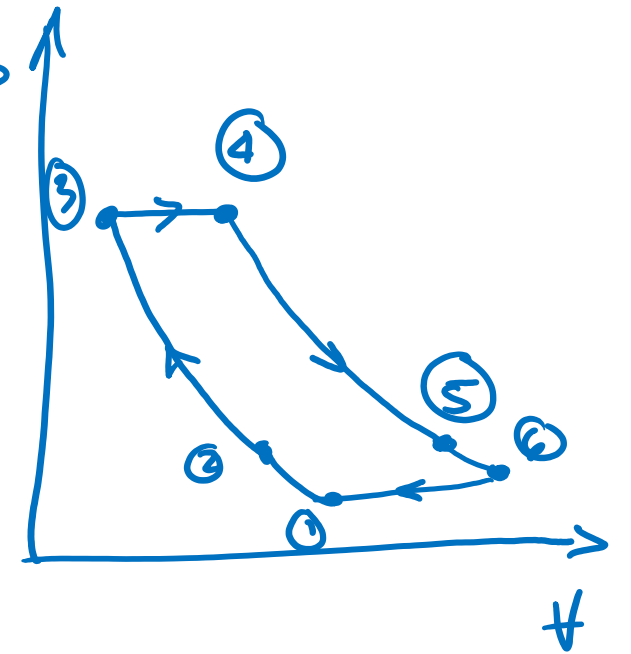
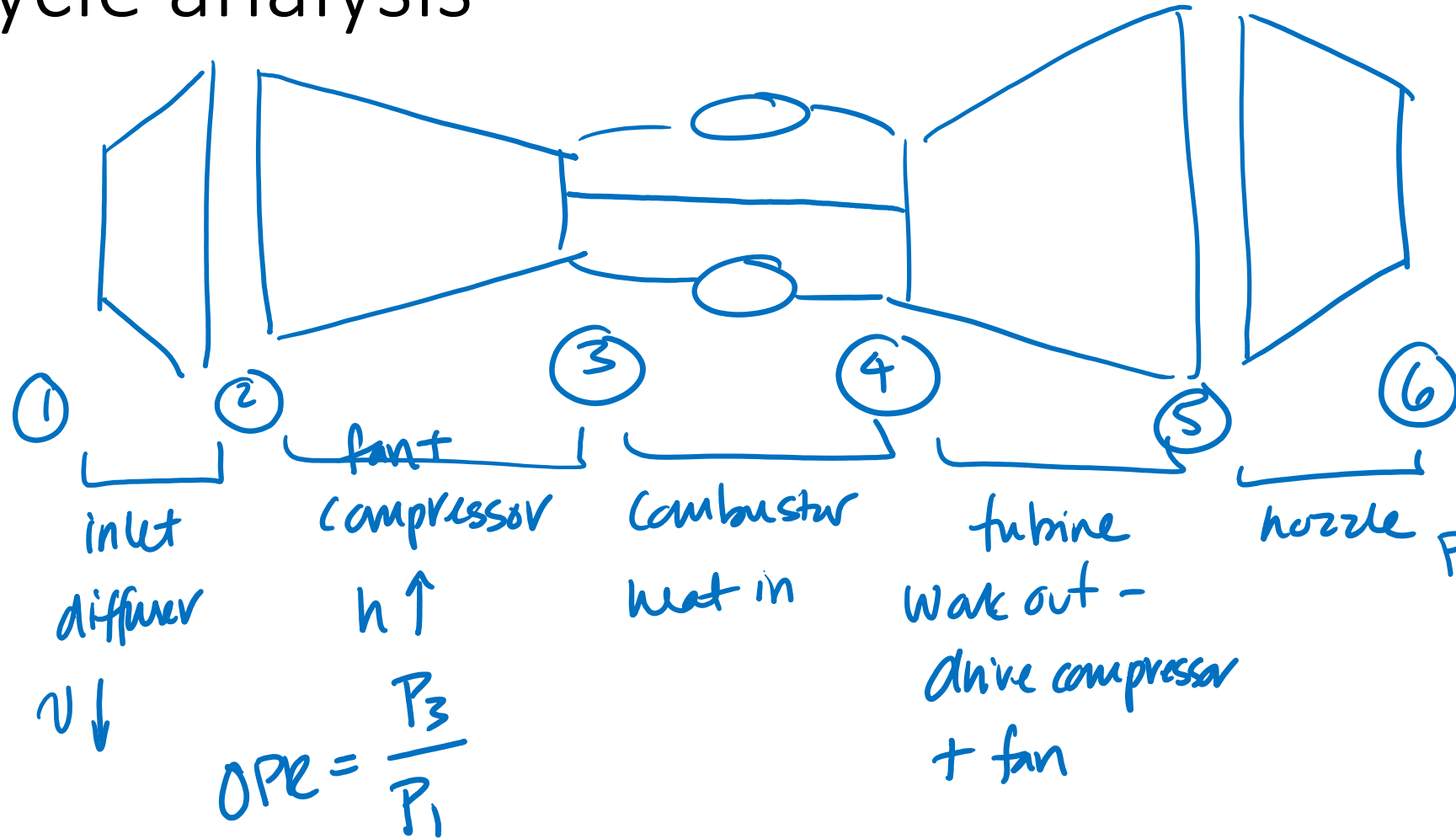
Courtesy of Pratt & Whitney

Augmented Turbofans

F100-PW-220/F100-PW-220E TURBOFAN ENGINE



Cycle analysis



Cycle analysis $\dot{m}(\Delta h + \Delta ke + \Delta pe) = \dot{Q} - \dot{W}$

1-2 : diffuser : $\Delta h + \Delta ke = 0$

5-6 : nozzle : $\Delta h + \Delta ke = 0$